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RESEARCH ARTICLE

Injectable Composite Hydrogels Orchestrating Macrophage Reprogramming and Chondrogenesis to Promote Microfracture-Assisted Cartilage Repair

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ABSTRACT

Articular cartilage repair remains a major clinical challenge. Although microfracture (MF) is widely applied, it frequently results in fibrocartilaginous repair with limited mechanical durability and unsatisfactory long-term outcomes. The persistent inflammatory microenvironment following cartilage injury disrupts tissue homeostasis and impairs chondrogenic differentiation of bone marrow stem cells (BMSCs), representing a major impediment to regeneration. Here, an injectable, thermosensitive composite hydrogel, constructed from a dopamine-modified hyaluronic acid and Pluronic F127 network, which incorporates chlorogenic acid (CA) and ZIF-8 nanoparticles encapsulated with kartogenin (KGN), is developed to establish a cascade repair strategy. Preferential release of CA can efficiently reprogram macrophages toward a pro-regenerative M2 phenotype and improve anti-inflammatory cytokine secretion, thereby creating a pro-regeneration microenvironment. Subsequently, the sustained release of KGN further stimulates BMSCs chondrogenic differentiation within this optimized niche. Biological assays demonstrate that this synergistic mechanism enhances cartilage-specific matrix synthesis and alleviates matrix degradation under inflammatory conditions. Furthermore, this composite hydrogel, combined with MF, improves cartilage tissue regeneration in a rat model, as evidenced by smooth defect filling, well-organized extracellular matrix deposition, and reduced Matrix metalloproteinase 13-mediated degradation. This work presents a synergistic immuno-chondroregenerative platform that overcomes fundamental limitations of MF and offers a promising paradigm for functional cartilage regeneration.

1 | Introduction

Articular cartilage defects caused by acute trauma, chronic wear, or specific pathological conditions (e.g., osteoarthritis, rheumatoid arthritis) have become one of the leading causes of disability worldwide [1]. Due to the absence of vascular supply, innervation, and lymphatic drainage within cartilage tissue, coupled with its homogeneous cellular composition, the intrinsic regenerative

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capacity of cartilage is severely limited, making complete regeneration challenging once injury occurs. Current clinical treatments include microfracture (MF), allograft transplantation, autologous chondrocyte implantation, and matrix-induced autologous chondrocyte implantation. Among these, MF has been widely adopted as a first-line treatment due to its minimally invasive nature, low cost, and favorable short-term outcomes [2, 3]. The MF technique involves drilling into the subchondral bone plate to facilitate the migration of blood and BMSCs into the defect area, forming a fibrin-rich provisional clot containing stem cells. This approach effectively alleviates pain in the short term [4]. However, the absence of a specific microenvironment conducive to chondrogenesis often leads BMSCs to differentiate into fibrocartilage, which exhibits significantly inferior mechanical properties and wear resistance compared to native hyaline cartilage [5, 6]. To enhance the efficacy of MF-mediated cartilage repair and regeneration, developing adjuvant therapeutic strategies that optimize the local microenvironment and precisely regulate BMSCs toward hyaline cartilage differentiation holds significant research potential.

Research has elucidated that macrophage polarization imbalance is a critical pathological factor hindering cartilage repair [7]. Macrophages, based on their functional state, can be categorized into resting (M0), pro-inflammatory (M1), and anti-inflammatory (M2) phenotypes. Under pathological conditions associated with cartilage injury, resident macrophages within the synovial tissue undergo phenotypic transformation, characterized primarily by a significant increase in M1 macrophages, leading to disruption of the M1/M2 macrophage dynamic equilibrium [8]. M1 macrophages, through the secretion of pro-inflammatory cytokines, reactive oxygen species (ROS), and enzymatic substances, induce sustained inflammatory responses [9]. Studies have demonstrated that the inflammatory microenvironment not only promotes chondrocyte transition or dedifferentiation toward a fibroblast-like phenotype, resulting in the formation of mechanically inferior fibrocartilage, but also disrupts cartilage homeostasis by inhibiting anabolic processes and promoting catabolic processes, thereby establishing a vicious cycle of persistent extracellular matrix (ECM) degradation [10]. Conversely, M2 macrophages, through the secretion of anti-inflammatory factors such as IL-10 and TGF-β, can effectively suppress inflammatory responses and promote tissue repair. Given the high plasticity of macrophages, targeted induction of M2 macrophage polarization via biomaterials has emerged as an effective strategy for optimizing the cartilage repair microenvironment.

Chlorogenic acid (CA), one of the most abundant polyphenolic compounds in the human diet, serves as a primary bioactive component in traditional Chinese medicinal herbs such as *Lonicera japonica* and *Eucommia ulmoides*. CA has been demonstrated to possess a wide range of biological activities, including antioxidant, anti-inflammatory, analgesic, antiviral, antitumor, and anti-ulcer properties [11]. In recent years, the regulatory effects of CA on macrophage M1/M2 polarization have been investigated. Li et al. reported a CA-based microneedle patch that promotes wound healing by reducing M1 macrophage polarization and significantly amplifying M2 repolarization [12]. Wang et al. demonstrated that CA-loaded nanocomposite hydrogels can regulate macrophage phenotype, inhibit the secretion of pro-inflammatory cytokines, increase the levels of the anti-inflammatory factor IL-10, and promote inflammation resolution [13]. However, current research on CA-mediated promotion of cartilage regeneration through regulation of macrophage M2 polarization remains unexplored. Based on existing evidence, we hypothesize that CA can inhibit local inflammation by inducing macrophage phenotypic transformation, thereby providing a stable regenerative microenvironment for cartilage repair.

On the basis of regulating the inflammatory microenvironment, continuously stimulating the differentiation of stem cells and the synthesis of the extracellular matrix is the core element to ensure effective cartilage regeneration. Kartogenin (KGN), a small-molecule compound, exhibits specific chondroinductive capacity for directing stem cell differentiation toward chondrocytes. Studies have shown that KGN can upregulate the expression of chondrogenic marker genes and proteins through multiple signaling axes, thereby promoting chondrocyte matrix regeneration [14, 15]. Xiao et al. developed a silk fibroin-based drug delivery scaffold capable of sustained release of KGN and the anti-inflammatory molecule, which effectively repaired full-thickness articular cartilage defects and improved joint motor function [16]. By combining the chondrogenic induction properties of KGN with the immunomodulatory functions of CA, constructing a cartilage repair scaffold with multiple biological activities is expected to provide an efficient treatment strategy for cartilage regeneration.

Addressing the two key challenges of an imbalanced inflammatory microenvironment and impaired chondrogenic differentiation of BMSCs in the cartilage defect area, this study proposed the construction of a smart hydrogel system capable of controlled sequential release of CA and KGN. The aim was to enhance cartilage repair outcomes following MF surgery by synergistically maintaining BMSC retention, modulating the inflammatory response, and directing BMSC chondrogenic differentiation. As illustrated in Figure 1a, KGN-loaded ZIF-8 nanoparticles (KGN@ZIF-8, KZ) were first prepared, and then co-dispersed with CA into a dopamine-modified hyaluronic acid (HA-DA) and Pluronic F-127 (F-127) precursor solution to construct an injectable, thermosensitive composite hydrogel (HDF/CA@KZ). Following MF surgery, the composite hydrogel was injected into articular cartilage defects, where its porous 3D architecture will provide abundant cell-anchoring sites, significantly enhancing BMSCs adhesion and retention to establish a stable in situ regenerative cell pool (Figure 1b). Subsequently, CA was preferentially released to modulate the M1/M2 macrophage polarization balance, suppressing the inflammatory response and establishing a pro-reparative microenvironment. Meanwhile, the ZIF-8 nanoparticles gradually degraded under the weakly acidic joint cavity environment (pH 6.5–6.8), and the sustained release of KGN specifically induced BMSC differentiation toward chondrocytes, promoting the synthesis of extracellular matrix components. This study provides an effective strategy for achieving functional regeneration of cartilage defects.

2 | Results and Discussion

2.1 | Synthesis and Characterization of KZ Nanoparticles

Metal–organic framework materials, such as ZIF-8 has high specific surface area, abundant pore structures, and excellent bio-

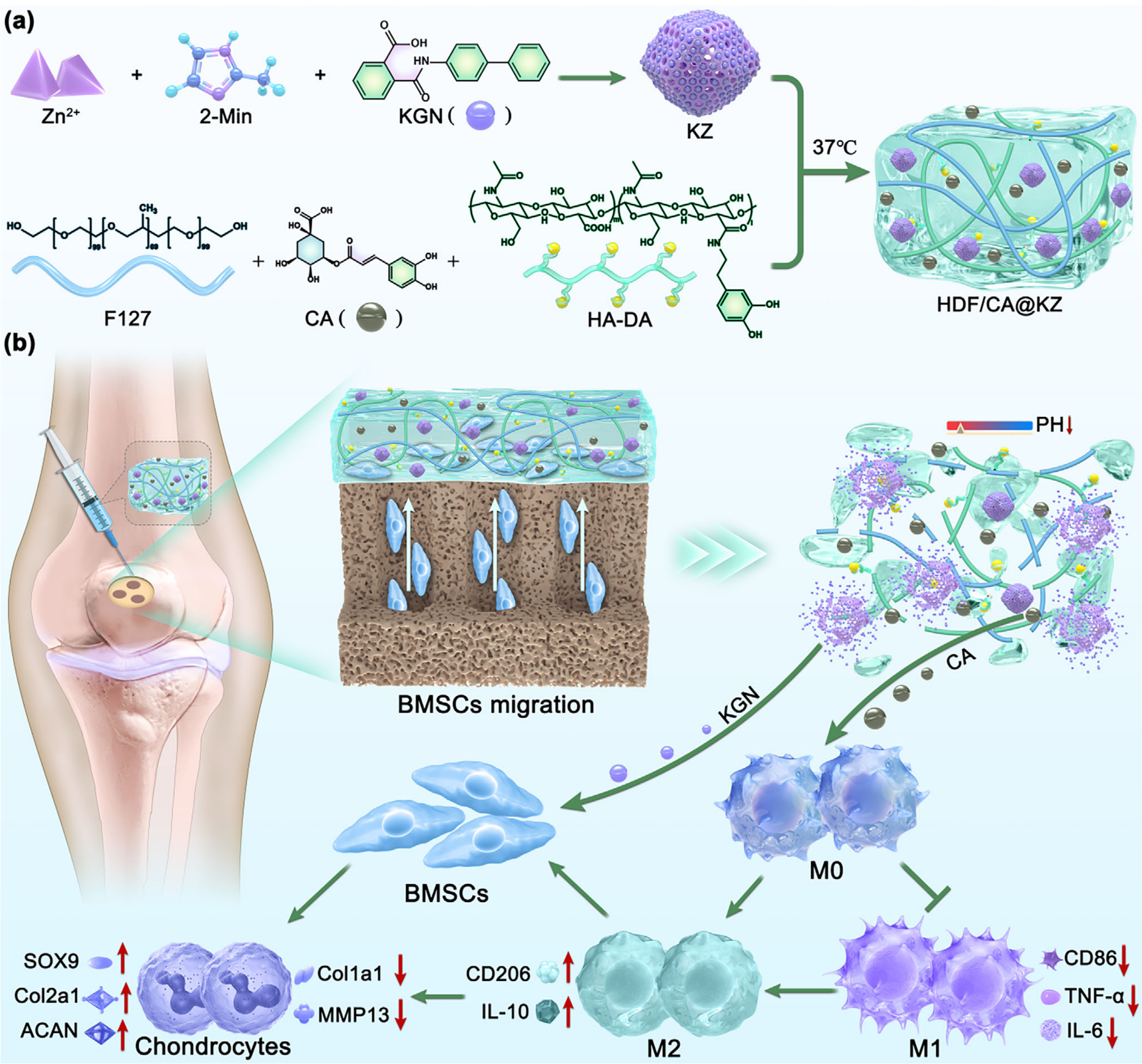

FIGURE 1 | (a) Synthesis of the HDF/CA@KZ hydrogel. (b) Schematic illustration of HDF/CA@KZ hydrogel for cartilage regeneration.

compatibility. They exhibit outstanding drug loading capacity and can achieve stimulus-responsive release through pH-triggered structural dissociation [17]. In this study, KGN was effectively incorporated into ZIF-8 via an organic solvent-mediated assembly process. Quantitative analysis revealed that the loading capacity of KGN within the ZIF-8 nanoparticles reached 26.5%. The transmission electron microscope (TEM) confirmed that the KZ nanoparticles had a distinct hexagonal crystal structure, with an average particle size of 200 nm (Figure 2a). This was slightly larger than that of the individual ZIF-8, whose particle size was approximately 100 nm. Nitrogen adsorption-desorption experiments confirmed the surface area and pore size distribution of ZIF-8 and KZ nanoparticles (Figure 2b,c). The test results indicated that ZIF-8 nanoparticles possessed micropores and mesopores, exhibiting a higher total adsorbed volume and likely a larger surface area and pore volume than KZ. These changes could be attributed to the encapsulated KGN, which caused partial blockage of the ZIF-8 pores [18]. These results collectively confirmed that KGN was successfully loaded into ZIF-8 nanoparticles.

2.2 | Fabrication and Characterization of HDF/CA@KZ Hydrogels

Hydrogels possess excellent biocompatibility, high water content, and low friction coefficient, and have been widely used as scaffold materials for cartilage repair [19]. As the main component of the extracellular matrix (ECM), HA demonstrates significant advantages due to its ability to promote cell migration, proliferation, and collagen accumulation [20, 21]. However, its inherent mechanical weaknesses and poor tissue adhesion limit its wider application. The catechol groups in DA can form strong interfacial adhesion with hydrated biological surfaces through cooperative covalent bonding and non-covalent interactions [22]. Here, we utilized

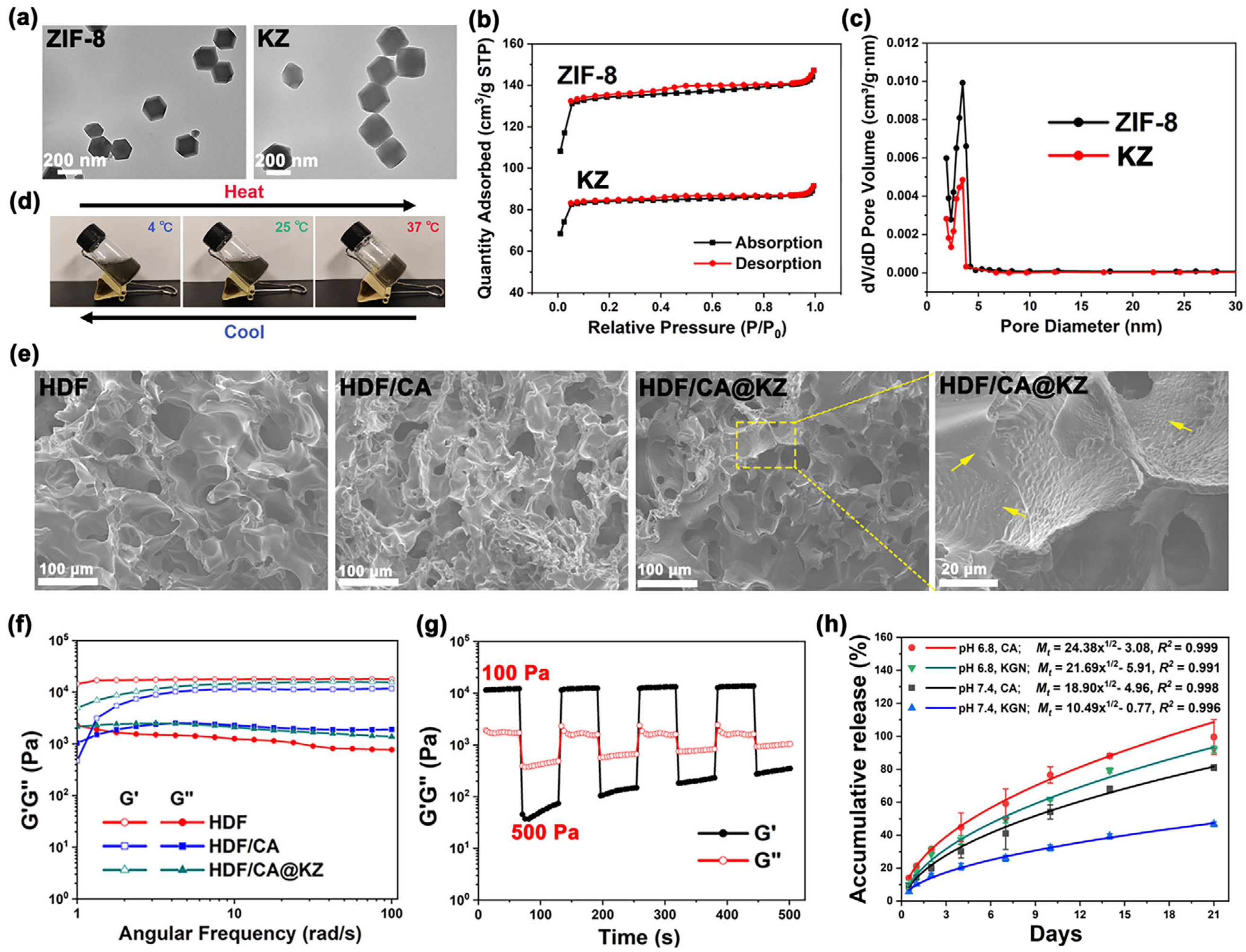

FIGURE 2 | Characterization of KZ nanoparticles and HDF/CA@KZ composite hydrogels. (a) TEM images of ZIF-8 and KZ. Scale bar: 200 nm. (b) N_2 adsorption-desorption isotherms and (c) corresponding pore size distributions of ZIF-8 and KZ. (d) Photographs showing the thermo-responsive sol-gel transition of the HDF/CA@KZ composite hydrogel at different temperatures. (e) SEM micrographs of HDF, HDF/CA, and HDF/CA@KZ hydrogels. Yellow arrows indicate the embedded KZ nanoparticles. Scale bar: 100 and 20 μm. (f) The storage modulus (G') and loss modulus (G'') of the HDF, HDF/CA, and HDF/CA@KZ hydrogels. (g) Self-healing behavior of the HDF/CA@KZ hydrogel evaluated by a continuous step-stress test. (h) In vitro cumulative release profiles of KGN and CA from the HDF/CA@KZ hydrogel under different pH conditions. Data are expressed as mean ± standard deviations (SD) ($n = 3$).

the adhesion property of DA to synthesize HA-DA conjugates to enhance the tissue adhesion performance of the HA-based Hydrogels. The successful synthesis of the HA-DA conjugate was confirmed by UV–vis spectroscopy. Native HA exhibited negligible absorbance in the 250–350 nm range, whereas the HA-DA conjugate showed a distinct absorption peak centered at approximately 280 nm, characteristic of the catechol moiety of dopamine (Figure S1a). The grafting degree of DA in HA-DA was determined using UV–vis absorbance analysis. According to the standard curve of DA (Figure S1b), the degree of substitution of HA by DA was calculated to be 6.09%, which was critical for the subsequent gelation behavior and antioxidant performance of the HDF hydrogel.

F127 is a poly(ethylene oxide)-poly(propylene oxide)-poly(ethylene oxide) (PEO-PPO-PEO) triblock copolymer with temperature-responsive sol-gel transition behavior [23]. Below the critical micelle temperature, F127 exists in a low-viscosity sol state, which is easy to be mixed with bioactive components and injected to achieve precise filling of irregular cartilage defects [24]. When the temperature rises to the physiological level, the hydrophobic effect of the PPO segment drives its rapid gelation, forming a stable three-dimensional network. To harness the structural adaptability of F127, here we engineered a composite hydrogel by combining HA-DA with F127. The temperature-sensitive characteristics of the HDF/CA@KZ hydrogel system could be observed in Figure 2d. At temperatures of 4°C or 25°C, the hydrogel remained in a fluid state, while at 37°C, it transitioned to a solid form. The co-assembly of HA-DA conjugates and F127 into a hydrogel was driven by π–π stacking interactions of HA-DA and hydrogen bonding between HA-DA and F127, facilitated by F127 forming micelles [22]. Moreover, the addition of CA enhanced the cross-linking of the hydrogel by providing additional hydrogen bonding and π–π stacking interactions, thereby strengthening the network structure. Figure 2f illustrates the rheological properties of the

HDF, HDF/CA, and HDF/CA@KZ hydrogels. For all hydrogel groups, the energy storage modulus (G′) and loss modulus (G″) remained stable over the frequency range of 1–100 rad/s. Notably, all hydrogels exhibited solid-like viscoelastic behavior, characterized by a higher G′ than G,″ indicating successful gelation [25]. Subsequently, the rheological recovery behavior of the HDF/CA@KZ hydrogels was investigated (Figure 2g). Under cyclic stress alternating between 100 and 500 Pa, the storage modulus G′ showed a sharp decrease when the applied stress reached 500 Pa, suggesting disruption of the elastic network. Upon reducing the stress back to 100 Pa, G′ rapidly recovered, demonstrating the reversible self-healing capability of the hydrogel. The highly reproducible "damage-recovery" behavior over multiple cycles further confirmed the stable and reversible viscoelastic properties of the HDF/CA@KZ hydrogels [26]. The scanning electron microscope (SEM) results showed that the freeze-dried HDF hydrogel exhibited an interconnected porous structure with a smooth surface (Figure 2e). While maintaining the inherent porosity of the original HDF matrix, the addition of CA led to a reduction in pore size, which might be attributed to the nanoscale reinforcement mechanism, thereby effectively increasing the cross-linking density of the polymer network. High-resolution imaging further confirmed the uniform dispersion of KZ particles in the HDF/CA@KZ hydrogel matrix without any obvious aggregation. The phase composition of the HDF/CA@KZ hydrogel matrix was determined by X-ray diffraction (XRD), which revealed diffraction peaks corresponding to both KZ and CA (Figure S2a).

The controlled biodegradation of hydrogels is critical for their application in drug delivery and tissue regeneration, as it directly influences the release profile of encapsulated drugs and the long-term stability of the implanted material. Here, the biodegradation characteristics of the composite hydrogels were further evaluated. As shown in Figure S2b, the HDF hydrogel exhibited a degradation rate of approximately 70%, whereas the HDF/CA and HDF/CA@KZ hydrogels demonstrated a more moderate degradation rate, with a mass loss of approximately 50%–60% within 30 days. The incorporation of CA into the hydrogel matrix resulted in a reduced degradation rate, likely due to the formation of multiple non-covalent interactions, including hydrogen bonds, π–π stacking interactions, and oxidative covalent bonds between the CA and HDF components, which enhanced the mechanical strength and stability of the composite hydrogel. The biodegradation results indicated that the HDF/CA@KZ hydrogels could offer prolonged mechanical support and a more controlled release profile.

Spatiotemporally controlled drug delivery strategies play a pivotal role in the multi-stage and synergistic treatment of cartilage defects, as different regenerative phases require distinct biochemical cues [16]. In this study, a dual-drug delivery platform was rationally designed to address the inflammatory and chondrogenic stages of cartilage regeneration. Specifically, CA, which exhibited well-documented anti-inflammatory activity, was directly encapsulated within the hydrogel matrix to enable relatively rapid release, thereby facilitating early suppression of the inflammatory cascade. In contrast, KGN, a small-molecule chondrogenic inducer, was loaded into ZIF-8 nanoparticles and subsequently incorporated into the hydrogel network to achieve sustained and prolonged release, supporting the continuous chondrogenic differentiation of BMSCs during the later regenerative phase. To investigate the drug release behavior under physiologically relevant conditions, the drug release behavior was conducted in PBS at pH 7.4, representing normal tissue environments, and pH 6.8, simulating the mildly acidic microenvironment associated with inflammation. As shown in Figure 2h, approximately 81% of the encapsulated CA was released from the HDF/CA@KZ hydrogels after 21 days at pH 7.4. In contrast, under acidic conditions (pH 6.8), CA exhibited a markedly accelerated release profile, reaching nearly complete release within the same time frame. This pH-dependent behavior was primarily attributed to enhanced hydrolytic degradation of the hydrogel network, which facilitated drug diffusion. The release kinetics of CA and KGN in early (0–7 days) and late (7–21 days) stages were further analyzed. In the early stage (Figure S3a), CA exhibited a higher release rate than KGN under both pH conditions, confirming preferential early release of CA. In the late stage (Figure S3b), the release rate of KGN increased while CA release gradually plateaued under pH 6.8 conditions. These quantitative results claimed that CA was preferentially released at early stages, followed by delayed KGN release. The delayed release of KGN was primarily due to its encapsulation in ZIF-8 nanoparticles, which acted as an additional diffusion barrier, whereas CA was directly loaded into the hydrogel matrix and diffuses more freely [27]. Moreover, the pH-responsive degradation of ZIF-8 further modulated KGN release (Figure S3c). Under weakly acidic conditions (pH 6.8), partial ZIF-8 degradation and slight hydrogel swelling accelerated KGN diffusion; whereas at neutral pH (7.4), slower ZIF-8 degradation and a denser hydrogel network retarded KGN release, resulting in more pronounced delayed release. Collectively, these results demonstrated that the HDF/CA@KZ hydrogel achieved programmed sequential release of CA and KGN, precisely matching the stage-specific requirements for cartilage repair.

2.3 | ROS-Scavenging Ability of HDF/CA@KZ Hydrogels

Cartilage injury is frequently accompanied by a pronounced inflammatory response, which leads to excessive accumulation of ROS in the local microenvironment and consequently impairs cartilage regeneration [28]. To evaluate the antioxidant capacity of the HDF/CA@KZ hydrogels, both DPPH free radical scavenging assays and intracellular ROS measurements were performed. As shown in Figure 3a, incubation of DPPH solution with HDF, HDF/CA, and HDF/CA@KZ hydrogels for 10 min resulted in a marked decrease in the characteristic absorption peak compared with the blank control. Notably, the CA-containing hydrogels almost completely abolished the DPPH absorption signal, indicating a strong radical scavenging capability. Quantitative analysis further revealed that the DPPH scavenging efficiency of the HDF hydrogel reached approximately 46%, whereas both the HDF/CA and HDF/CA@KZ hydrogels achieved nearly complete scavenging (Figure 3b). In addition, the DPPH chromogenic reaction of the HDF/CA@KZ hydrogel exhibited a clear time-dependent decrease in absorbance at 517 nm, confirming sustained antioxidant activity over time (Figure 3c). These findings substantiated the effective antioxidative performance of the hydrogel complex. The intrinsic antioxidant activity of the HDF hydrogel could be primarily attributed to the redox-active catechol moieties derived

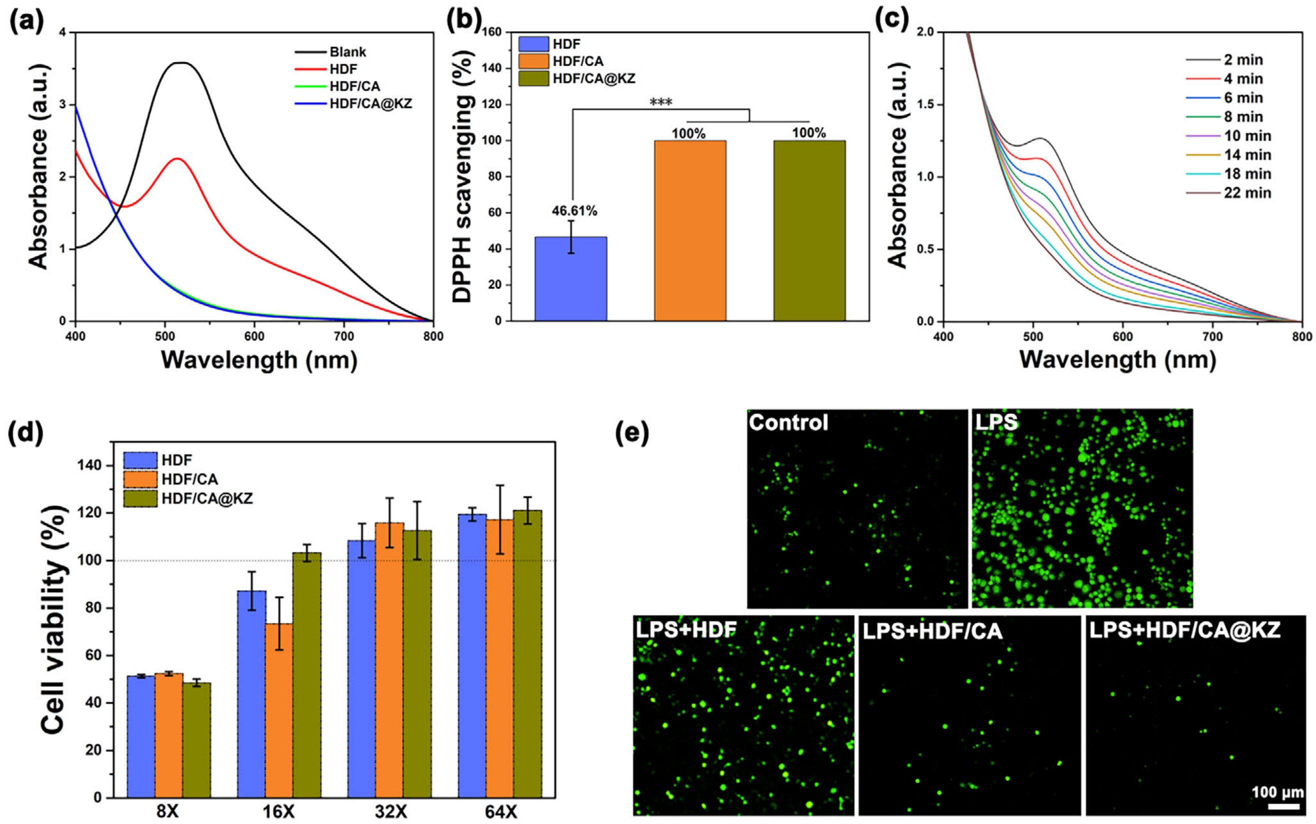

FIGURE 3 | ROS-scavenging ability of HDF/CA@KZ hydrogels. (a) UV–vis absorbance spectra of DPPH solutions after treatment with extracts from Blank, HDF, HDF/CA, and HDF/CA@KZ hydrogels. (b) Quantification of the DPPH radical scavenging efficiency of HDF, HDF/CA, and HDF/CA@KZ hydrogel extracts. Data are expressed as mean ± SD ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). (c) Time-dependent UV–vis absorbance spectra of DPPH solution after incubation with HDF/CA@KZ hydrogel extract. (d) Cell viability of RAW 264.7 cells cultured with different dilutions (8X, 16X, 32X, and 64X) of hydrogel extracts. Data are expressed as mean ± SD ($n = 5$). (e) Fluorescence micrographs showing intracellular ROS levels in LPS-stimulated RAW 264.7 macrophages following treatment with different hydrogel extracts. Scale bar: 100 μm.

from the DA component, which were capable of donating hydrogen atoms or electrons to neutralize free radicals, consistent with previous reports [29, 30]. The significantly enhanced scavenging performance observed in the CA-containing hydrogels further highlighted the contribution of CA, a well-established natural antioxidant. Owing to its abundant phenolic hydroxyl groups and strong redox activity, CA effectively eliminated ROS and other oxidative species, thereby synergistically enhancing the overall antioxidant performance of the composite hydrogel system [31]. Such robust ROS-scavenging capability was expected to mitigate oxidative stress in the inflammatory microenvironment, which was beneficial for protecting resident cells and facilitating subsequent cartilage regeneration.

Before assessing intracellular ROS levels, the cytocompatibility of the hydrogels was first evaluated using an MTT assay, with RAW264.7 macrophages as the target cells. As shown in Figure 3d, cell viability was measured in the presence various hydrogel at different dilutions (8×, 16×, 32×, and 64×). At the 8× dilution, cell viability across all hydrogel groups was relatively low, approximately 50%–60%. However, when the hydrogel products were diluted 32×, cell viability in all groups significantly improved, with viability exceeding 100%. The estimated concentrations of CA and KZ in the 32× diluted extracts were approximately 15.6 and 31.3 μg/mL, respectively. In this study, unless stated otherwise, all experiments involving RAW264.7 macrophages as the research subjects were conducted using a final concentration corresponding to a 32× dilution for each hydrogel group. It was also noteworthy that a 32× dilution was used in the DPPH free radical scavenging assay for consistency across experimental conditions.

To assess ROS generation in RAW264.7 cells exposed to different hydrogel substrates, the intracellular ROS levels were measured using the fluorescent probe 2′,7′-dichlorofluorescein diacetate (DCFH-DA). As shown in Figure 3e, the control group exhibited baseline ROS levels, while LPS-stimulated cells demonstrated a significant increase in oxidative stress. Lipopolysaccharide (LPS), a well-known endotoxin, activates macrophages and induces ROS production [32, 33]. Notably, cells cultured with HDF hydrogels showed a slight reduction in fluorescence intensity compared to the LPS-treated control. More significantly, cells exposed to the extracts from CA-incorporated hydrogels (HDF/CA and HDF/CA@KZ) exhibited a marked attenuation of ROS-related fluorescence signals. Excessive ROS can trigger harmful cell cascades, leading to the degradation of important extracellular matrix components, triggering chondrocyte apoptosis, and ultimately disrupting the delicate balance necessary for effective cartilage repair and regeneration [34, 35]. These findings were consistent with the DPPH scavenging test results, suggesting that

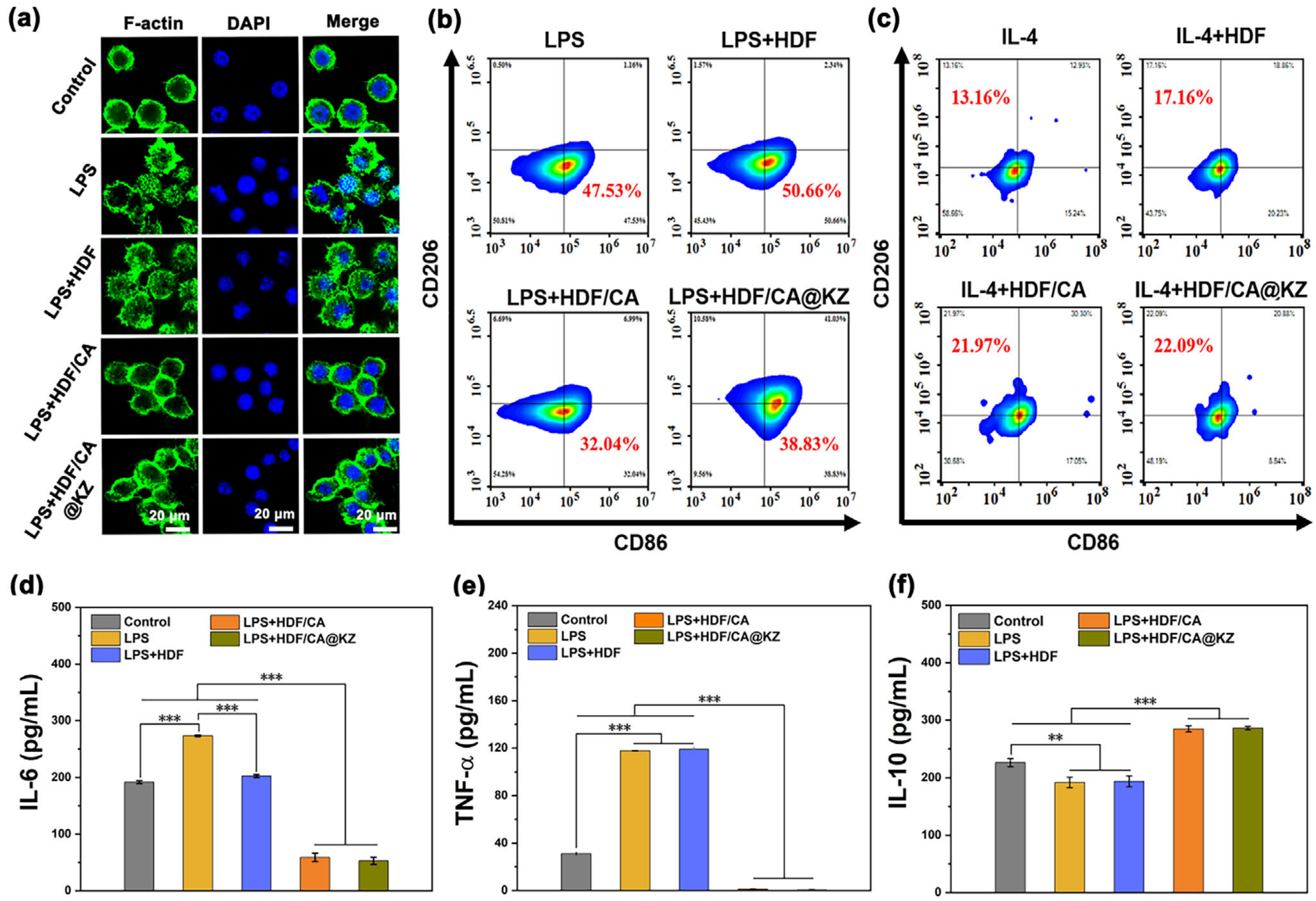

FIGURE 4 | Immunomodulatory effects of the composite hydrogels on macrophage polarization in vitro. (a) Immunofluorescence staining of F-actin and nuclei in LPS-stimulated RAW 264.7 macrophages co-cultured with hydrogel extracts. Scale bar: 20 μm. (b, c) Flow cytometry analysis of macrophage polarization markers (CD86 for M1, CD206 for M2) after treatment with hydrogel extracts under LPS or IL-4 stimulation. (d–f) Quantification of pro-inflammatory (IL-6, TNF-α) and anti-inflammatory (IL-10) cytokines in culture supernatants by ELISA. Data are expressed as mean ± SD ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$).

the developed composite hydrogels could relieve oxidative stress in inflammatory environments, which possessed considerable potential for supporting the retention of the natural functions of cartilage.

2.4 | Effects of HDF/CA@KZ Hydrogels on Inflammation Through Immunomodulation

The immunoregulatory dynamics of tissue-resident immune cells critically orchestrate regenerative processes, with macrophage phenotypic determination emerging as a pivotal regulatory node [36–38]. To evaluate the influence of the HDF/CA@KZ hydrogel on macrophage phenotype modulation, RAW264.7 macrophages were cultured with hydrogel extracts under LPS stimulation, and corresponding morphological changes were examined. As shown in Figure 4a, LPS stimulation induced a pronounced morphological transformation of RAW264.7 cells from a rounded shape to a flattened morphology with extensive pseudopodia extension, which is characteristic of classically activated (M1) macrophages. In contrast, following incubation with the CA-incorporated hydrogel extract, RAW264.7 macrophages exhibited complete pseudopodia retraction accompanied by bipolar elongation, assuming a morphological characteristic correlating with M2-like polarization state. These observations demonstrated that the HDF/CA@KZ hydrogel could effectively suppress LPS-induced M1 polarization while promoting macrophage polarization toward an M2-like phenotype.

To further verify the transformation of the macrophage phenotype to an anti-inflammatory activation state, immunofluorescence staining was performed to evaluate the expression of CD86, a marker associated with classically activated (M1) macrophages, and CD206, a hallmark of alternatively activated (M2) macrophages (Figure S4). It was found that the fluorescence intensity associated with CD86 (M1) was markedly lower in the HDF/CA and HDF/CA@KZ groups compared to the LPS and HDF groups. Conversely, the fluorescence intensity of CD206 (M2) in these groups was significantly higher than in the other groups, indicating that the incorporation of CA promoted macrophage polarization toward an M2-like phenotype. Furthermore, flow cytometry analysis of RAW264.7 cells also confirmed these results. The percentage of CD86-positive cells was significantly reduced in the CA-incorporated hydrogels, decreasing from 50.66% in the LPS control group and 47.53% in the HDF group to 32.04% and 38.83% in the

HDF/CA and HDF/CA@KZ groups, respectively (Figure 4b). The percentage of CD206-positive cells in macrophages treated with the hydrogel containing CA significantly increased, exceeding the level of the IL-4 stimulation group (Figure 4c), indicating a strong M2 polarization effect. In addition, the functional polarization of macrophages was also evaluated by using ELISA to quantify cytokine secretion. As illustrated in Figure 4d,e, the expression levels of pro-inflammatory cytokines (IL-6, TNF-α) were significantly elevated after LPS treatment. However, following treatment with the hydrogel containing CA, their expression levels were significantly reduced compared to both the LPS group and the control group. In contrast, the levels of anti-inflammatory cytokines, particularly IL-10 (Figure 4f), were significantly elevated in the HDF/CA and HDF/CA@KZ groups.

Previous studies have verified that Zn^{2+} ions can drive macrophage polarization from the pro-inflammatory M1 to anti-inflammatory M2 phenotype and inhibit pro-inflammatory cytokine secretion [39, 40]. In this study, the possible influence of Zn^{2+} ions released by ZIF-8 on macrophage polarization was further examined. Flow cytometry analyses (Figure S5) showed LPS stimulation alone resulted in a high proportion of M1 pro-inflammatory macrophages (52.56%), while HDF/ZIF-8 treatment only slightly reduced M1 polarization to 49.48%. In sharp contrast, HDF/CA@ZIF-8 treatment significantly decreased the M1 population to 36.52%. Consistently, Figure S6 showed LPS markedly upregulated pro-inflammatory cytokines (IL-6, TNF-α); this upregulation was unmitigated by HDF or HDF/ZIF-8 alone but notably suppressed by HDF/CA@ZIF-8. These results demonstrated that ZIF-8-released Zn^{2+} exerted negligible immunomodulatory effects, whereas CA was the dominant effector inhibiting pro-inflammatory responses. This insignificance may be due to local Zn^{2+} concentrations below the effective threshold or sustained release differing from routine acute exposure in previous literature.

These comprehensive data collectively indicated that the HDF/CA@KZ hydrogel could effectively inhibit the polarization of macrophages toward the pro-inflammatory M1 phenotype, while promoting the transformation toward the anti-inflammatory and regenerative M2 phenotype. Chronic inflammation is a significant characteristic of degenerative cartilage diseases, where continuous M1-type macrophage activity exacerbates tissue damage as it secretes catabolic cytokines (such as IL-6, TNF-α), which degrades extracellular matrix (ECM) components and inhibits the function of chondrocytes [10]. By actively polarizing macrophages toward the M2-dominated phenotype, the hydrogel loading CA establishes a regenerative immune microenvironment. M2-type macrophages not only could promote the resolution of inflammation but also could secrete a series of nutritional and synthetic factors, which support the proliferation of chondrocytes, enhance the synthesis of extracellular matrix, and promote tissue remodeling [41]. In summary, by regulating the polarization of macrophages to simultaneously weaken destructive inflammatory signals and enhance reparative signals, the HDF/CA@KZ hydrogel provides a highly promising strategy for promoting functional cartilage repair.

2.5 | Effects of HDF/CA@KZ Hydrogels on Chondrogenic Differentiation and Chondroprotection In Vitro

To further elucidate whether macrophages modulated by the HDF/CA@KZ hydrogels could influence the chondrogenic potential of BMSCs, a macrophage-conditioned culture system was established to investigate the paracrine crosstalk between RAW264.7 macrophages and BMSCs (Figure 5a). Prior to investigation, a systematic assessment of the cell compatibility of the HDF/CA@KZ hydrogel with BMSCs was carried out. BMSCs were cultured with hydrogel extracts at different dilution ratios, and cell viability was evaluated using the MTT assay (Figure S7). The results showed that when exposed to a 32× hydrogel extract, the survival rates of all cell groups exceeded 100%, indicating support for cell proliferation. Live/dead staining further confirmed this, with predominantly green fluorescence and few dead cells observed for the 32× hydrogel extracts (Figure S8). This cell compatibility feature was consistent with the results obtained in RAW264.7 macrophages, suggesting that the HDF/CA@KZ hydrogel had an appropriate therapeutic safety window.

To assess the chondrogenic response of BMSCs to different hydrogel formulations, cells were cultured in macrophage-conditioned media (CM) derived from the respective treatment groups. After 7 days of culture, the morphological changes of BMSCs were observed by F-actin staining. As illustrated in Figure 5b, the BMSCs in the control, LPS, and HDF hydrogel groups presented a polygonal shape. In contrast, the BMSCs exposed to CM derived from the HDF/CA group displayed partial morphological changes, including cell rounding and the formation of multicellular aggregates, suggesting an early tendency toward chondrogenesis. This phenotypic transformation could be attributed to the immunomodulatory properties of CA, which may counteract the inhibitory effect of the inflammatory microenvironment on chondrogenesis. It was noteworthy that the BMSCs cultivated in the HDF/CA@KZ group showed more obvious chondrocyte-like characteristics, including extensive cell rounding, increased cell volume, and early signs of extracellular matrix deposition. Furthermore, the expression of chondrogenic markers, including Col2a1, ACAN, and SOX9, was quantified by ELISA. As shown in Figure 5c–e, exposure to an inflammatory microenvironment markedly suppressed the expression of these chondrogenic markers in the LPS group compared with the control. In contrast, BMSCs cultured in conditioned media from the HDF/CA group exhibited partially restored expression of ACAN, SOX9, and Col2a1, whereas a more pronounced upregulation was observed in the HDF/CA@KZ group, indicating enhanced chondrogenic differentiation. These observations indicated that the addition of KGN could synergistically promote chondrogenic differentiation when combined with the immunomodulatory signals provided by the hydrogel containing CA. Recently, the chondrogenic bioactivity of Zn^{2+} ions have been verified [42]. Here, the influence of Zn^{2+} ions released by ZIF-8 degradation on chondrogenic differentiation of BMSCs was further evaluated (Figure S9). The secretion levels of SOX9, ACAN, and Col2a1 showed no significant differences among the Control, HDF, HDF/CA, and HDF/CA@ZIF-8 groups. In contrast, the HDF/CA@KZ group presented significantly elevated SOX9 and Col2a1 secretion compared with the HDF/CA@ZIF-8 and Control

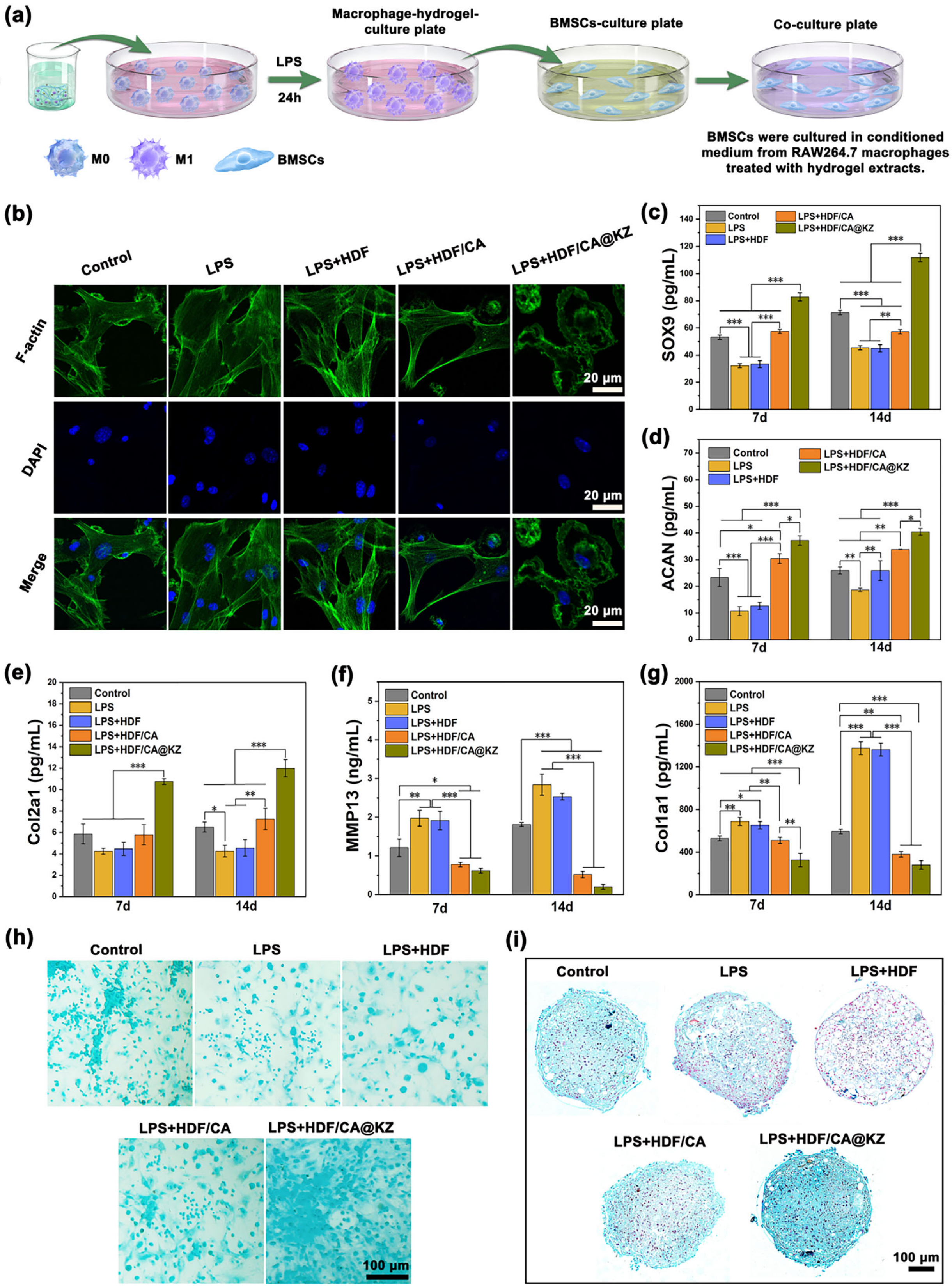

FIGURE 5 | Chondro-protective and anti-catabolic effects of the composite hydrogels on BMSCs in an inflammatory microenvironment simulated by macrophage-conditioned medium (CM). (a) Schematic illustration of the indirect co-culture model. (b) Immunofluorescence staining of F-actin and nuclei in BMSCs after culture with different CMs. Scale bar: 20 μm. ELISA quantification of the secretion of (c) SOX9, (d) ACAN, (e) Col2a1, (f) MMP13, and (g) Col1a1 by BMSCs after culture in different CMs for 7 and 14 days. Data are expressed as mean ± SD ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). (h) Alcian Blue staining of BMSC monolayer cultures after 14-day chondrogenic induction in different treatment group. Scale bar: 100 μm. (i) Histological Alcian Blue staining of chondrogenic BMSC pellet cross-sections after 21-day chondrogenic induction in different treatment group. Scale bar: 100 μm.

groups. These results demonstrated that Zn^{2+} ions released from ZIF-8 played a negligible role in BMSC chondrogenesis, and that KGN was the primary driver of enhanced chondrogenic differentiation. Given that excessive Zn^{2+} ions may transiently suppress chondrogenic differentiation, the regulatory role of Zn^{2+} ions is highly system- and concentration-dependent [43]. We therefore hypothesized that the locally released Zn^{2+} ions in our system failed to reach the effective concentration threshold required to activate pro-chondrogenic signaling pathways.

The inflammatory signals associated with cartilage damage will stimulate the expression of matrix metalloproteinases (MMPs), which, through protein hydrolysis and cleavage of key structural components, synergistically promote the degradation of the extracellular matrix and accelerate the catabolism of cartilage [36]. It is worth noting that MMP-13 is a key member of the collagenase family and plays a crucial regulatory role in the progression of osteoarthritis. Its expression level is positively correlated with the degree of cartilage degeneration [44, 45]. To further evaluate the matrix quality and phenotypic stability of the newly formed cartilage, the expression of MMP13 and Col1a1 was assessed. As shown in Figure 5f,g, compared with the control group and the LPS group, the expression of these two markers was significantly reduced in the HDF/CA and HDF/CA@KZ groups. The inhibition of MMP13 expression indicated that matrix degradation was suppressed and the preservation of cartilage matrix was improved. The reduction in Col1a1 expression indicated that the engineered cartilage successfully maintained the differentiation phenotype of chondrocytes and promoted the formation of a matrix similar to hyaline cartilage, rather than fibrotic repair [46, 47]. In summary, these results indicated that the HDF/CA@KZ hydrogel not only enhanced the differentiation of chondrocytes under inflammatory conditions, but also promoted the deposition of functional and metabolically inactive cartilage matrix.

To more intuitively demonstrate the promoting effect of HDF/CA@KZ hydrogel on chondrocyte differentiation and cartilage matrix formation, we performed Alcian blue staining at both the cellular level and in chondrocyte differentiation spheres to evaluate the deposition of sulfated glycosaminoglycans (GAG) (Figure 5h,i). Quantitative analysis was presented in Figure S10. The LPS treatment significantly reduced matrix deposition compared with the control group. In the HDF/CA group, staining intensity was markedly increased relative to the LPS group, indicating that the anti-inflammatory effect of CA partially alleviated GAG degradation induced by inflammation. Notably, the HDF/CA@KZ group exhibited the most intense and extensive Alcian Blue staining, reflecting significantly higher GAG accumulation than all other groups. This enhanced matrix deposition was likely attributable to the sustained release of KGN from the composite hydrogel, which has been reported to activate the SOX9-mediated chondrogenic signaling pathway, thereby promoting proteoglycan synthesis and regulating chondrocyte metabolic activity [48]. Overall, the above research results indicated that the HDF/CA@KZ hydrogel could effectively promote the differentiation of BMSCs into cartilage. The coordinated upregulation of cartilage differentiation markers (Col2a1, ACAN, and SOX9), as well as the inhibition of matrix-degrading enzymes (MMP13) and fibrosis markers (Col1a1), highlighted the dual functions of the prepared HDF/CA@KZ hydrogel in cartilage protection and the formation of transparent-like cartilage matrix.

2.6 | Evaluation of HDF/CA@KZ Hydrogels on Cartilage Repair In Vivo

Although MF remains a widely used clinical strategy for the treatment of cartilage defects, its regenerative outcomes are often limited, particularly with respect to the formation of durable hyaline cartilage [49, 50]. Integrating hydrogel technology with MF procedures might potentially address surgical constraints, as hydrogel matrices facilitate the localized retention of BMSCs while directing tissue repair progression [5, 51]. To assess cartilage repair efficacy, here we established a rat knee model with full-thickness cartilage defects (2 mm in diameter × 2 mm in depth), comparing MF monotherapy with HDF/CA@KZ hydrogel combination therapy. No adverse tissue reactions were observed throughout the 4-week treatment period, demonstrating that the hydrogel was highly biocompatible and well-tolerated in vivo. At 4 weeks post-implantation, knee joints were harvested for macroscopic examination, histological analysis, and immunohistochemical evaluation. As shown in Figure 6a, macroscopic assessment revealed incomplete defect filling and irregular cartilage surfaces in the control and HDF hydrogel groups. The HDF/CA group exhibited modest improvement in defect coverage compared with the control and HDF groups. In contrast, defects treated with the HDF/CA@KZ hydrogel displayed substantially improved repair, with the defect area largely filled by cartilage-like tissue exhibiting a relatively smooth surface that approached the appearance of surrounding native cartilage, although a slight surface depression remained.

The histological analysis conducted through H&E staining further confirmed that the HDF/CA@KZ hydrogel exhibited outstanding regenerative capabilities. As shown in Figure 6b, in the control group and the HDF group, there was a significant amount of fibrous tissue deposition at the injury site. The HDF/CA hydrogel group showed that the defect was filled with tissue similar to cartilage, but there were still obvious boundaries within the repair area. In contrast, the HDF/CA@KZ hydrogel group achieved the most ideal repair effect, with the defect completely surrounded by new cartilage tissue. The regenerated cartilage reached the same thickness as the surrounding natural tissue and had a healthy appearance similar to natural cartilage. The magnified images further revealed that there were a large number of chondrocytes distributed in the regenerated tissue. Furthermore, the cartilage-specific matrix was characterized through Safranin O-Fast Green (SO-FG) staining and Col2a1 immunohistochemical analysis. Figure 6c suggested a more pronounced deposition of ECM within defects treated with HDF/CA@KZ hydrogels when compared to the other experimental groups. It was also observed that the regenerated cartilage in the HDF/CA@KZ group presented distinct transverse markings at its junction with the subchondral bone. Moreover, the highest deposition of Col2a1 in the HDF/CA@KZ group, followed by the HDF/CA group (Figure 6d). To further evaluate the degradative activity and ECM stability of regenerated cartilage tissue, immunofluorescence staining for MMP13 was performed. As shown in Figure 6e, the regenerated tissue in the control group exhibited a strong red fluorescent signal, indicating elevated MMP13 expression

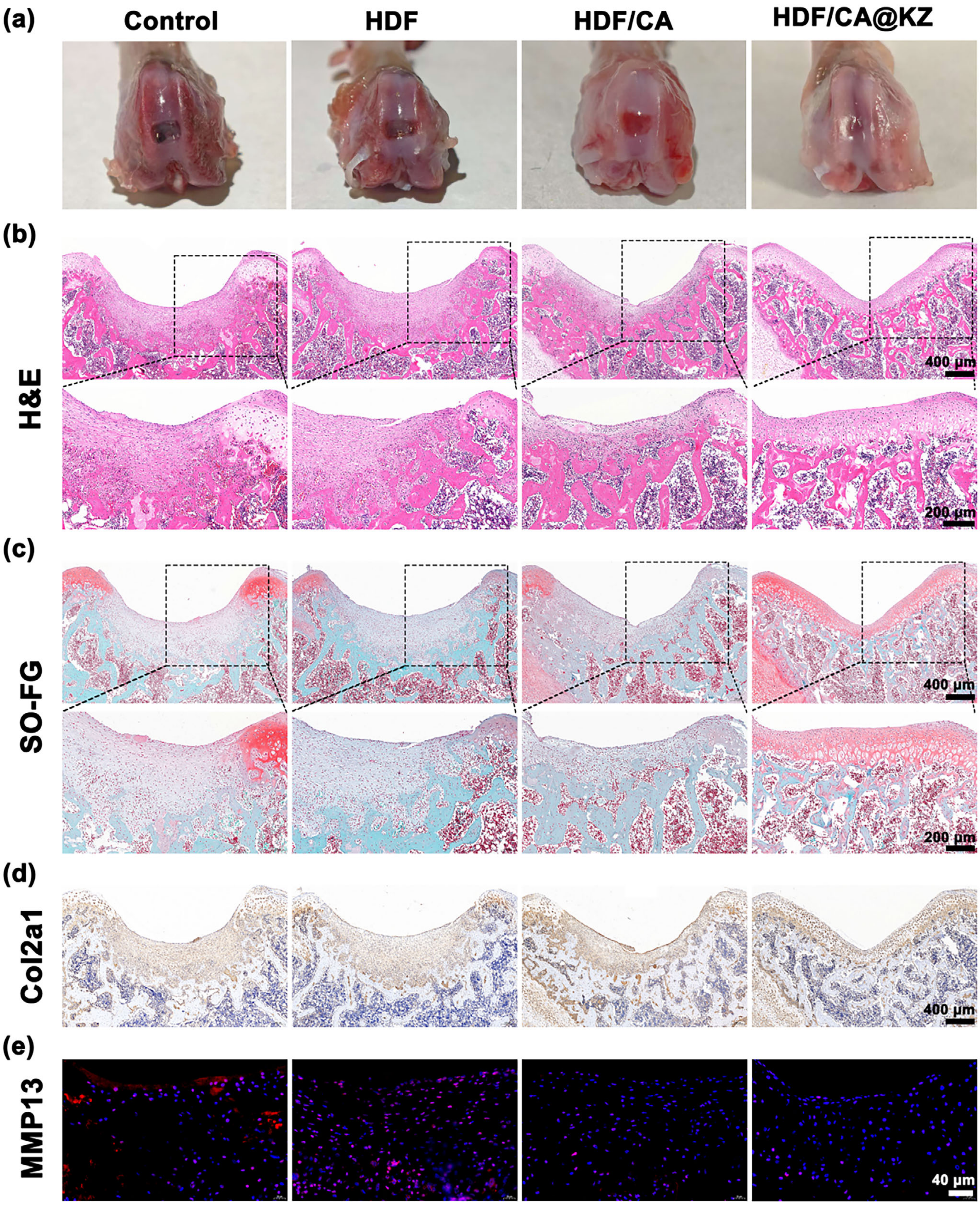

FIGURE 6 | In vivo evaluation of cartilage repair in a rat osteochondral defect model at 4 weeks post-surgery. (a) Macroscopic photographs of the femoral condyles showing the articular cartilage surface in different treatment groups. (b) H&E staining of histological sections from the defect sites. Scale bar: 400 μm. The lower panels are magnified views of the dashed box regions in the upper panels. Scale bar: 200 μm. (c) SO-FG staining of the defect sections. Scale bar: 400 μm. The lower panels are magnified views of the dashed box regions in the upper panels. Scale bar: 200 μm. (d) Immunohistochemical staining for Col2a1 in the regenerated tissue. Scale bar: 400 μm. (e) Immunofluorescence staining for MMP13 in the defect region. Scale bar: 40 μm.

levels and increased ECM degradation activity. In contrast, the red fluorescence intensity in the HDF/CA and HDF/CA@KZ groups was reduced compared to the control group, suggesting that the presence of CA effectively inhibited the activation of cartilage-degrading enzymes.

Collectively, all the above data indicated that combining MF with the HDF/CA@KZ hydrogel system could effectively overcome the limitations of MF treatment alone. The enhancement of the repair effect by the HDF/CA@KZ hydrogel could be summarized into three aspects. First, the hydrogel matrix provided effective structural support, helping to retain BMSCs locally, thereby guiding their directional differentiation into chondrocytes. Second, CA played a significant role in alleviating cartilage degeneration, as evidenced by the decreased expression of MMP13. Elevated MMP13 levels are recognized as a factor contributing to cartilage degeneration, which impairs the integrity of newly formed tissues [52, 53]. The added CA created a more favorable microenvironment for the maturation and integration of the newly synthesized extracellular matrix. Third, the continuous release of KGN further amplified the regenerative effect by enhancing the proliferation of chondrocytes and the synthesis of extracellular matrix. In summary, the HDF/CA@KZ hydrogel system effectively enhances MF-based cartilage repair, presenting considerable advantages for clinical implementation.

3 | Conclusion

This study designed a novel HDF/CA@KZ composite hydrogel, which combined ZIF-8 nanoparticles loaded with KGN and CA-functionalized bioadhesive hydrogel matrix, achieving controlled drug delivery and possessing both structural support and anti-inflammatory activity. The HDF/CA@KZ hydrogel demonstrated strong ROS scavenging ability and effectively promoted the polarization of macrophages toward the M2 phenotype. Additionally, it significantly enhanced chondrogenic differentiation of BMSCs and extracellular matrix synthesis in an inflammatory environment. In vivo experiments further confirmed that the combination of HDF/CA@KZ hydrogel with MF technology significantly improved cartilage repair, as evidenced by good new cartilage formation, orderly extracellular matrix deposition, and reduced MMP13-mediated degradation. In conclusion, the developed HDF/CA@KZ hydrogel provides a promising new approach for minimally invasive treatment of cartilage defects.

4 | Experimental Section

4.1 | Materials

Pluronic F-127 and zinc nitrate hexahydrate ($Zn(NO_3)_2{\cdot}6H_2O$) were purchased from Sigma–Aldrich. Dopamine hydrochloride was obtained from Energy Chemical. 1-(3-Dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (EDC), N-hydroxysuccinimide (NHS), and HA (molecular weight 40–100 kDa) were supplied by Macklin. KGN and 2-methylimidazole were purchased from Aladdin. CA was obtained from Aladdin. All reagents were of analytical grade and used as received without further purification.

4.2 | Synthesis of KZ

Briefly, 360 mg of $Zn(NO_3)_2{\cdot}6H_2O$ and 792 mg of 2-methylimidazole, along with 2 mg of KGN, were separately dissolved in 12 and 24 mL of methanol, respectively. The two solutions were then mixed and stirred at room temperature for 45 min. Following this, the mixture was centrifuged (9000 rpm, 15 min) and washed three times with methanol. Finally, the precipitates were collected and lyophilized to obtain KZ nanoparticles.

4.3 | Preparation of HDF/CA@KZ Composite Hydrogels

Pluronic F-127 solution was first prepared by dissolving 2 g of F-127 in 5 mL of deionized water at 4°C under gentle stirring until complete dissolution. HA-DA was synthesized via carbodiimide-mediated coupling. Briefly, 1 g of HA was dissolved in 100 mL of deionized water, and the pH was adjusted to 5.0–6.0 using 0.1 M HCl. Subsequently, dopamine hydrochloride (569 mg), EDC (575 mg), and NHS (345 mg) were added sequentially, and the reaction was allowed to proceed for 24 h under a nitrogen atmosphere to prevent dopamine oxidation. The reaction mixture was dialyzed against acidified deionized water (pH 4.5) using a dialysis membrane (MWCO 10 kDa) for 3 days, followed by lyophilization to obtain HA-DA. The grafting degree of DA onto HA was determined by measuring the absorbance at 280 nm with a UV–vis spectrophotometer (INFINITE EPLEX, Tecan, Switzerland) and quantitatively determined with a DA standard. To prepare the hydrogel precursor, HA-DA (0.1 g) was dissolved in 5 mL of deionized water and mixed with the F-127 solution at a volume ratio of 2:1 to form the HDF hydrogel. CA and KZ nanoparticles were subsequently added to the HDF precursor at final concentrations of 500 μg/mL and 1 mg/mL, respectively, yielding the HDF/CA@KZ composite hydrogel.

4.4 | Material Characterization

The morphology of KZ nanoparticles was characterized using TEM (HT7800, Hitachi). Specific surface area and pore size distribution were determined by nitrogen adsorption–desorption analysis using a BET surface analyzer (Micromeritics 3Flex). The microstructure of freeze-dried hydrogels was examined by SEM (SU-8010, Hitachi). Phase composition of the hydrogels was analyzed by XRD (Bruker D8 Advance/Ultima IV). Rheological properties were evaluated using a rotational rheometer (MARS60, Thermo Scientific) equipped with parallel plate geometry at 25°C. The storage modulus (G') and loss modulus (G'') were recorded under oscillatory shear. The sol–gel transition temperature of the hydrogels was determined using a vial inversion (tilt) test in a water bath.

4.5 | In Vitro Degradation of Composite Hydrogels

For degradation studies, lyophilized hydrogel samples (initial dry weight 0.1 g) were immersed in 10 mL of phosphate-buffered saline (PBS, pH 7.4) and incubated at 37°C under static conditions. The PBS was refreshed weekly to maintain sink conditions. At predetermined time points, samples were collected, lyophilized, and weighed to determine mass loss. Mass loss percentage was calculated as: Mass lose $(\%) = \frac{m_0 - m_t}{m_0} \times 100\%$, Where m_0 was the initial dry weight and m_t was the dry weight at time t.

4.6 | In Vitro Release of CA and KGN

The release profiles of CA and KGN from the HDF/CA@KZ hydrogels were evaluated at pH 7.4 and pH 6.5. Hydrogel samples were incubated in PBS at 37°C, and at predetermined time points, the supernatant was collected and replaced with fresh PBS. The concentration of CA and KGN was quantified using UV–vis spectroscopy based on a standard calibration curve. The release profiles of KGN from the KZ nanoparticles under the same pH conditions were evaluated using the same method.

4.7 | Cell Culture

BMSCs were isolated from bilateral femurs and tibiae of 4-week-old rats and expanded in α-MEM medium supplemented with 10% fetal bovine serum (FBS, Cellmax, China) and 1% penicillin/streptomycin (Beyotime). RAW264.7 macrophages were purchased from Wuhan Shangen Biotechnology Co., Ltd. After resuscitation, the RAW264.7 macrophages were cultured in RPMI-1640 medium supplemented with 10% FBS and 1% penicillin/streptomycin. All cells were maintained at 37°C in a humidified atmosphere containing 5% CO_2.

4.8 | Biocompatibility of Hydrogels

The cytocompatibility of the hydrogels was assessed using MTT assays and live/dead staining. The HDF/CA@KZ hydrogel extract was prepared by incubating 1.2 g of HDF hydrogel in 5 mL of basal culture medium at 37°C for 14 days. After incubation, CA and KZ nanoparticles were added to achieve final concentrations of 500 µg/mL and 1 mg/mL, respectively, yielding a 1× extract. This 1× extract was then diluted with complete medium to generate 8×, 16×, 32×, and 64× working concentrations. For MTT assays, BMSCs or RAW264.7 cells were seeded in 96-well plates at a density of 5×10^4 cells/mL and cultured overnight. The culture medium was then replaced with hydrogel extracts of different dilutions and incubated for an additional 24 h. Cell viability was evaluated using the MTT reagent following standard protocols. For live/dead staining, BMSCs were treated with 32× hydrogel extracts for 24 h, then stained with Calcein AM and Propidium Iodide (PI) and imaged by fluorescence microscopy (Nikon, EVOS M7000).

4.9 | ROS Scavenging Properties of Hydrogels

The free radical scavenging activity of the hydrogels was evaluated using a DPPH assay. Briefly, 32× diluted hydrogel extracts were mixed with 3.0 mL of 400 µM DPPH methanol solution and incubated in the dark for 10 min. Absorbance was measured at 517 nm using a UV–vis spectrophotometer. The DPPH scavenging efficiency was calculated according to the following equation: *DPPH clearance rate* $(\%) = \frac{A_B - A_S}{A_B} \times 100\%$. Where A_B represents the absorbance of the blank and A_S represents the absorbance of the sample. Intracellular ROS levels were assessed using DCFH-DA. RAW264.7 cells were seeded in 24-well plates and incubated for 24 h, followed by treatment with 32× hydrogel extracts in the presence of 100 ng/mL LPS for an additional 24 h. Cells were then incubated with DCFH-DA at 37°C for 30 min, and fluorescence images were acquired using a fluorescence microscope.

4.10 | Macrophage Polarization Modulation by Hydrogels

To assess macrophage polarization, RAW264.7 cells were cultured with 32× hydrogel extracts in 12-well plates. Polarization was induced by supplementation with either 100 ng/mL LPS for M1 induction or 40 ng/mL IL-4 for M2 induction. Macrophage phenotypes were quantified by flow cytometry (Agilent, USA) using FITC-conjugated anti-CD86 (M1 marker) and APC-conjugated anti-CD206 (M2 marker) antibodies. Immunofluorescence staining with the same antibodies, followed by Hoechst 33342 nuclear staining, was performed and visualized using fluorescence microscope. Cytoskeletal organization was examined by staining F-actin with TRITC-conjugated phalloidin and nuclei with DAPI, followed by photographed by confocal microscopy (ZEISS Celldiscoverer 7). The inflammatory response was further quantified by ELISA. RAW264.7 cells were seeded in 24-well plates and treated with hydrogel extracts and LPS (100 ng/mL). After 24 h, culture supernatants were collected, and levels of IL-6, TNF-α, and IL-10 were measured using ELISA kits according to the manufacturers' instructions.

4.11 | Evaluation of Chondrogenic Differentiation under Inflammatory Conditions

BMSCs were seeded in 24-well culture plates at a density of 5×10^4 cells per well. To evaluate chondrogenic differentiation under inflammatory conditions, a conditioned medium-based co-culture model was established. RAW264.7 macrophages were incubated with 32× diluted hydrogel extracts and stimulated with LPS to induce M1 polarization. After 24 h of incubation, the culture supernatants were collected and used as macrophage-conditioned media. Chondrogenic induction was performed using high-glucose Dulbecco's modified Eagle's medium supplemented with 1% ITS+ Premix, 50 µg/mL ascorbic acid, 100 µg/mL sodium pyruvate, 100 nM dexamethasone, and 40 µg/mL L-proline. During subsequent biological assays, BMSCs were cultured in a 1:1 (v/v) mixture of chondrogenic induction medium and macrophage-conditioned medium. After 7 days of induction, cytoskeletal organization was assessed by fixing cells with 4% paraformaldehyde and staining F-actin with rhodamine-phalloidin. Fluorescence images were then acquired using a confocal microscopy. At 7 and 14 days, the expression levels of chondrogenic markers, including Col2a1, ACAN, and SOX9, as well as the matrix-degrading enzyme MMP13 and the fibrotic marker Col1a1, were quantified using ELISA. After 21 days of culture, sulfated glycosaminoglycan deposition was evaluated by Alcian Blue staining in both monolayer cultures and spheroid constructs. The optical images were photographed by a confocal microscopy, and the images were analyzed by Image J software.

4.12 | In Vivo Cartilage Defect Repair

Animal experiment was performed in accordance with institutional and national guidelines, and the experimental procedures were approved by the Animal Experimentation Ethics Committee of Shandong First Medical University (Approval No. W202402290093). Cartilage defect model was established using male adult Sprague–Dawley (SD) rats weighing 200–250 g. Following anesthesia, the knee joint was surgically exposed, and a cylindrical cartilage defect (with a diameter of 2 mm and a thickness of 2 mm) was created in the femoral trochlea using a trephine drill. The animals were randomly assigned to four groups: MF group, HDF+MF group, HDF/CA+MF group, and HDF/CA@KZ+MF group. MF was performed by drilling a 1 mm Kirschner wire into the center of the defect until fluid exudation from the bone marrow cavity was observed. Following surgery, the rats were allowed unrestricted activity and were provided standard food and water. Animals were sacrificed at 4 weeks postoperatively. The knee joints were harvested, photographed, and fixed in paraformaldehyde. Subsequently, the specimens were decalcified with EDTA and embedded in paraffin for histological analyses. H&E and Safranin O-Fast Green staining were performed according to standard protocols to assess the quality of the newly regenerated cartilage tissues. Furthermore, the expression of Col2a1 and MMP13 in the repaired cartilage area was analyzed by immunohistochemical staining and immunofluorescence staining, respectively.

4.13 | Statistical Analysis

Each experiment was repeated at least three times. A minimum of three samples were used for statistical analysis for each test. Statistical analysis was performed using GraphPad Prism 8.0 software. One-way analysis of variance with Tukey's post hoc test was used for multiple group comparisons, and two-tailed unpaired Student's *t*-test was applied for two-group comparisons. The results were expressed as the mean ± standard deviation (SD). Statistical significances were expressed as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adhm71374-sup-0001-SuppMat.docx.